

Harnessing the Generative Quantum Eigensolver for Next-Generation Materials Design

Global Industry Challenge (GIC) 2026

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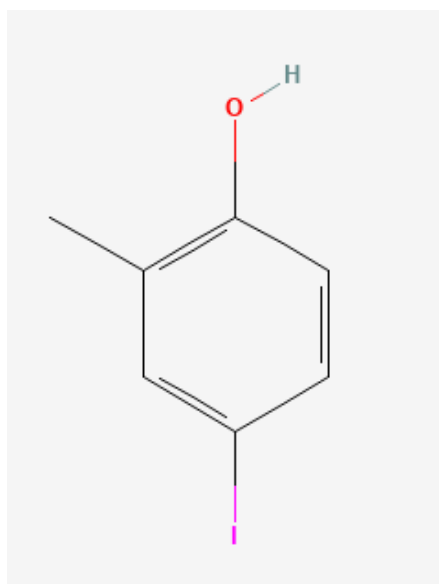


Figure 1: 4-iodo-2-methylphenol (IMePh) structure. Source: PubChem.

Team Name

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Strategic Context

This Phase 1 concept addresses the core quantum-simulation bottleneck in EUV photoresist design: predicting the photoabsorption spectrum and the downstream electron-emission pathways that control lithographic blur. The solution is designed for FTQCs, not near-term NISQ devices, because the relevant observables require deeper circuits, richer excited-state structure, and more stable error budgets than can be expected from shallow prototypes.

Proposed Solution

The proposed solution is a hybrid quantum-classical Generative Quantum Eigensolver (GQE) workflow: a generative model proposes compact circuit structures, and a hybrid training loop ranks them against target energy and transition metrics for 4-iodo-2-methylphenol (IMePh). It is designed to recover the EUV photoabsorption spectrum, low-lying excited states, and transition observables while remaining systematic enough to extend across chemically related molecules through materials-informatics transfer. Our deliverable will be a reproducible prototype that computes prioritized quantum-derived metrics for IMePh and related molecules, then returns a ranked candidate list for further study¹.

Quantum Modeling Workflow

This workflow follows the EUV photophysics chain from 92 eV photoionization through Auger decay and secondary-electron generation. Candidate Hamiltonians are evaluated with differentiable quantum circuits, and the results are used to rank molecules by predicted Auger electron yield and related emission behavior.

We model each candidate with a compact electronic Hamiltonian, map it to qubits using Jordan–Wigner, and evaluate operator sequences from a learned pool,

$$\rho(\mathbf{j}) = U_{j_L} \cdots U_{j_1} \rho_0 U_{j_1}^\dagger \cdots U_{j_L}^\dagger, \quad E(\mathbf{j}) = \text{Tr}[\hat{H}\rho(\mathbf{j})],$$

The GQE workflow then optimizes a sequence-level objective that biases lower-energy circuits,

$$\mathcal{L}_{\text{GQE}} = \left(\sum_{i=1}^L w_{j_i} - E(\mathbf{j}) \right)^2.$$

For spectroscopy we estimate Auger-sensitive observables through the response,

$$\Gamma(\omega) \propto \sum_{F \neq I} \frac{|\langle F | \hat{A} | I \rangle|^2 \eta}{(E_F - E_I - \omega)^2 + \eta^2},$$

which enables direct comparison to classical Auger reference spectra and emission-line trends.

Technical Stack and Implementation Path

PennyLane is the implementation layer for Hamiltonian construction, differentiable circuits, shot management, and device-agnostic execution on simulators or FTQC-oriented backends. A classical generative model proposes ansatz sequences, and the GQE loop ranks them with measured observables, giving a reproducible path from the notebooks to a small family of EUV-relevant compounds.

Conclusion

This proposal couples GQE-style search with a PennyLane workflow for FTQC-scale materials discovery. Validation focuses on prioritized metrics and ranking quality against classical references on tractable active spaces, with the emphasis on feasibility, reproducibility, and decision utility.

¹Primary reference: K. Keithley et al., "Auger Spectroscopy via Generative Quantum Eigensolver: A Quantum Approach to Molecular Excitations," <https://arxiv.org/abs/2603.12859v1>